<u>Note</u> By relative volume comparison if

$$(1-\delta)\frac{Vol(B_{1/2R}(p))}{Vol(B_{1/2R}(\underline{0}))} \leq \frac{Vol(B_R(p))}{Vol(B_R(\underline{0}))} \quad (*)$$

then $\forall \eta \leq 1/2$ we have

$$\left(1-\Psi(\delta|\eta,n,R)\right)\frac{Vol(B_{R'}(p))}{Vol(B_{R'}(\underline{0}))} \leq \frac{Vol(\partial B_{R'}(p))}{Vol(\partial B_{R'}(\underline{0}))}$$

for $R' = (1-\eta)R$

$\Rightarrow$ conclusion of Thm holds under $(*)$.

<u>Theorem</u> Let $Ric_{M^n} \geq -(n-1)\delta$

$$Vol(B_R(p)) \geq (1-\delta)Vol(B_R(0)) \quad 0 \in \mathbb{R}^n$$

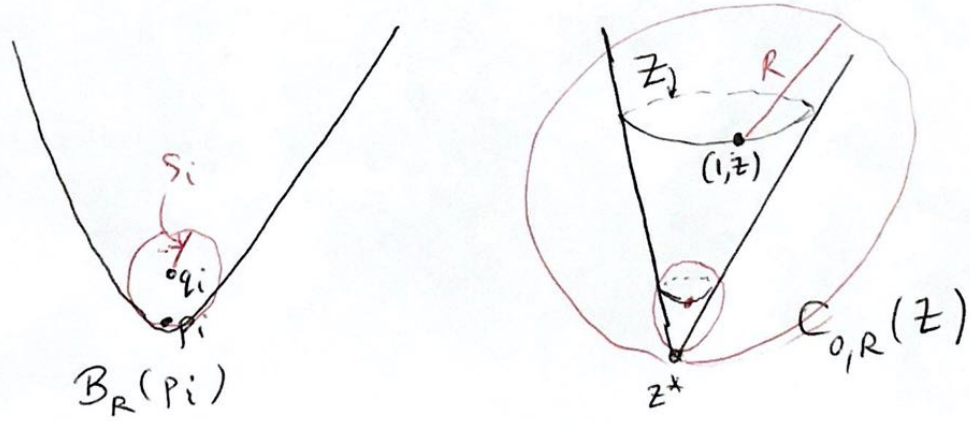
$$\Rightarrow \quad d_{GH}(B_R(p), B_R(0)) < \Psi(\delta|n,R)$$

<u>Proof</u> Suppose result fails. Then $\exists M_i^n$, $B_R(p_i) \subset M_i^n$ s.t. $Ric_{M_i^n} \geq -(n-1)\frac{1}{i}$

$$Vol(B_R(p_i)) \to Vol(B_R(0))$$

$$B_R(p_i) \xrightarrow{G.H.} C_{0,R}(Z) \neq \mathbb{R}^n$$

for some Z.



Let $z \in Z$. $\exists q_i \in B_{s_i}(p_i)$, $s_i \to 0$

s.t. $B_{s_i R}(q_i)$ scaled up by s_i^{-1}

converge to $B_R((1,z)) \subset C(Z)$

$$\mathrm{Vol}(\underbrace{B_{R+s_i}(q_i)}_{\supset B_R(p_i)}) \geq (1 - \Psi(i^{-1}))\, \mathrm{Vol}(B_{R+s_i}(0))$$

$\Rightarrow$ by vol. comparison

$$\mathrm{Vol}(B_{s_i R}(q_i)) \geq (1 - \Psi(i^{-1}))\, \mathrm{Vol}(B_{s_i R}(0))$$

By almost volume cone thm

$$\underbrace{B_R((1,z))}_{\subset C(Z)} = C_{0,R}(Z') \quad \text{for some } Z'$$

$\Rightarrow$ $C(Z)$ is a cone based at all $(1,z) \in C(Z)$

$\Rightarrow$ ray from z^* through $(1,z)$ extends as line.

By (almost) splitting thm, $C(Z) = \mathbb{R}^n$ $\square$

<u>Structure of limit spaces</u>

Let (M_i^n, m_i) be pointed Riem. n-folds s.t.

$$\mathrm{Ric}_{M_i^n} \geq -(n-1)$$

$$\mathrm{Vol}(B_1(m_i)) \geq \nu > 0 \quad \text{(non-collapsing)}$$

$$(M_i^n, m_i) \longrightarrow (Y, \bar{y}) \text{ in pointed GH sense.}$$

Let $y \in Y$, $r_i \to 0$

$(Y, y, r_i^{-1} d_Y)$ has subsequence converging in pointed GH sense to $(Y_y, y_\infty, d_\infty)$

<u>Defn.</u> Y_y is a tangent cone at y.

<u>Note</u>: $(M^n_{k(j)}, p_{k(j)}, r^{-1}_{k(j)} \rho_{k(j)}) \longrightarrow (Y_y, y_\infty, d_\infty)$

for suitable sequence $k(j) \to \infty$, ρ_i = dist. on M_i

Same holds for all iterated tangent cones.

<u>Almost splitting thm</u>: If an iterated tangent cone Y_{y_k} contains a line, then $Y_{y_k} = \mathbb{R} \times Y'$

<u>Almost vol. cone thm</u>: Every iterated tangent cone is a metric cone $C(X)$, diam $X \leq \pi$

<u>Defn.</u> for $\varepsilon < 1$ define ε-regular set $\mathcal{R}_\varepsilon \subset Y$ to be $y \in Y$ s.t. $\forall$ tangent cones (Y_y, y_∞) at y

$$d_{GH}(B_1(y_\infty), B_1(0)) < \varepsilon \quad , \quad 0 \in \mathbb{R}^n$$

$$\mathcal{R} = \bigcap_{\varepsilon > 0} \mathcal{R}_\varepsilon .$$

<u>Thm</u>
- $\forall \varepsilon > 0 \ \exists \delta > 0$ s.t. $\mathcal{R}_\delta \subset \mathring{\mathcal{R}}_\varepsilon$ (interior)

$$\Longrightarrow \quad \mathcal{R} = \bigcap_{\varepsilon > 0} \mathring{\mathcal{R}}_\varepsilon$$

- $\mathring{\mathcal{R}}_\varepsilon$ is bi-Hölder equiv. to smooth Riemannian mfld.

Hausdorff measure

Defn. Let $A \subset W$ in metric space W.

For $\ell, \eta > 0$, let

$$\mathcal{H}^\ell_\eta(A) = \inf_{\{B_{r_i}(w_i)\}} \omega_\ell \sum_i r_i^\ell$$

$= \text{vol of } B_1 \subset \mathbb{R}^\ell$ if $\ell \in \mathbb{Z}$

inf over all covers $A \subset \bigcup B_{r_i}(w_i)$ s.t. $r_i \le \eta$

$$\mathcal{H}^\ell(A) = \lim_{\eta \to 0} \mathcal{H}^\ell_\eta(A)$$

- $a \in A$ is a pt. of ℓ-density, $a \in D_\ell(A)$ if $\limsup_{r \to 0} \dfrac{\mathcal{H}^\ell_\infty(A \cap B_r(a))}{\omega_\ell r^\ell} \geq 2^{-\ell}$

  Then $\mathcal{H}^\ell(A \setminus D_\ell(A)) = 0$

- If $\mathcal{H}^\ell(A) < \infty$, then for $\mathcal{H}^\ell$-a.e. $a \in A$

$$\limsup_{r \to 0} \frac{\mathcal{H}^\ell(A \cap B_r(a))}{\omega_\ell r^\ell} \le 1$$

Stratification of singular set

Defn. Singular set $S = Y \setminus R$ and for $0 \le k \le n-1$ let $S_k \subset S$ be

$$S_k = \left\{ y \in Y \;\middle|\; \begin{array}{l}\text{no tangent cone } Y_y \text{ at } y \\ \text{is of form } \mathbb{R}^{k+1} \times Y'\end{array} \right\}$$

If one tangent cone at y is $\mathbb{R}^n$, then all are

$$\Rightarrow S = S_{n-1}$$

<u>Thm</u>

- $\dim S_k \leq k$
- $S_{n-1} = S_{n-2} \implies \dim S \leq n-2$

<u>Proof sketch</u>